

Grade: / 150

FDSFE Final Instructions

Due Date: 11:59pm Saturday December 6th, 2025

To submit, share this notebook to fortino@udel.edu and give edit permissions.

Resources

This exam is entirely open book, open note, and open internet. You may use any resource to help you on this exam *except* for other people or generative AI.

Getting Started

In order to start working on this final you must go to **File -> Save a copy in Drive**. Then, you can edit and rename the new copy of this notebook which you will submit.

Naming Conventions

Your submission notebook must follow specific name conventions:

- The name of this file is **FDSFE_F25_Final.ipynb**.
- When you make a copy of it, the new file will be named **Copy of FDSFE_F25_Final.ipynb**.
- You will rename this file in the following fashion:
 - First you will include the first letter of your first name, capitalized
 - Next you will include your surname, also capitalized.
 - Finally you will include an underscore followed by the original name of the assignment.

For example: My first name is Willow and my last name is Fortino, so I would rename the first assignment as **WFORTINO_FDSFE_F25_Final.ipynb**.

AI Use and Collaboration Statement

You are not allowed to use generative AI or large language models like chatGPT or Google Gemini on this exam. If you do, you will receive a 0 on this exam.

You are not allowed to work with others on this exam. If you do, you will receive a 0 on this exam.

Signature

Write your name here to affirm that you have read the AI Use and Collaboration statement.

[YOUR NAME HERE]

```
import numpy as np
from matplotlib import pyplot as plt
```

```
import keras
import sklearn
```

```
(x_train, y_train), (x_test, y_test) = keras.datasets.mnist.load_data()
```

Part 1: MNIST (90 points total)

Introduction

We will use the [MNIST dataset](#) to train a classification model.

Objective

Using the MNIST dataset, train a classifier to differentiate between pictures of even digits (0, 2, 4, 6, and 8) and odd digits (1, 3, 5, 7, 9). Your classifier must be a neural network. It can either be an ANN or a CNN.

Tasks

- (10 points) **Task 1a: Data Exploration and Visualization**
 - In the cell below, complete the `plot_samples` function. This function plots nine samples of the MNIST dataset.
 - The figure that `plot_samples` creates must adhere at all previously established standards for good figures.
- (10 points) **Task 1b: Figure Caption**
 - In the text cell below titled "Figure Caption", write a caption for your figure.
 - Explain what is being plotted.
 - Explain why the figure is important.
- (5 points) **Task 2: Preprocessing Inputs**
 - In the cell below, complete the `preprocess_inputs` function. This function accepts `x_train` or `x_test` and performs min-max scaling to between 0 and 1.
- (5 points) **Task 3: Preprocessing Labels**
 - The labels (`y_train` and `y_test`) label each image as a digit between 0 and 9. We need our labels to be a 0 if the sample is an image of an even digit or a 1 if the sample is an image of an odd digit.
 - In the cell below, complete the `preprocess_labels` function. The labels 0, 2, 4, 6, and 8 should be converted to a 0. The labels 1, 3, 5, 7, and 9 should be converted to a 1.
- (10 points) **Task 4: Neural Network**
 - In the cell below, complete the `make_model` function.
 - This function accepts `input_shape` which is the shape of the inputs.
 - Using Keras, create a neural network model. You may create a simple feed-forward ANN or a CNN.
 - Specify an optimizer, loss function, metric and learning rate, then compile the model.
 - This function returns `model` a Keras model object.
- (10 points) **Task 5: Training**
 - In the cell below, complete the `train_model` function. This function trains the model.
 - Specify an appropriate number of epochs and an appropriate batch size
 - Reserve a portion of the training set as a validation set during training.
- (10 points) **Task 6a: Loss and Metric Curves**
 - In the cell below, complete the `plot_curves` function. This function accepts the history dictionary of a keras model after training.
 - This figure should be a one column, two row plot.
 - The horizontal axis of the figure should be the training epochs.
 - The vertical axis of the figure should be the Loss (for the top row plot) and the Accuracy (for the bottom row plot).
- (10 points) **Task 6b: Figure Caption**
 - In the text cell below titled "Figure Caption", write a caption for your figure.
 - Explain what is being plotted.
 - Explain why the figure is important.
- (10 points) **Task 7a: Confusion Matrix**
 - In the cell below, complete the `plot_confusion_matrix` function. This function accepts the model inputs, the model labels and the model itself. The function calculates the confusion matrix and then plots it.

- When calculating the confusion matrix with `sklearn.metrics.confusion_matrix` use the `normalize='true'` keyword argument.
- (10 points) **Task 7b: Figure Caption**
 - In the text cell below titled "Figure Caption", write a caption for your figure.
 - Explain what is being plotted.
 - Explain why the figure is important.

Task 1a: Data Exploration and Visualization

```
def plot_samples(x):
```

```
    return fig
```

```
fig = plot_samples(x_train)
fig.show()
```

Task 1b: Figure Caption

[Double-click this cell to write your figure caption in this cell.]

Task 2: Preprocessing Inputs

```
def preprocess_inputs(x):
```

```
    return x_scaled
```

```
x_train_scaled = preprocess_inputs(x_train)
x_test_scaled = preprocess_inputs(x_test)
```

Task 3: Preprocessing Labels

```
def preprocess_labels(y):
```

```
    return y_binarized
```

```
y_train_01 = preprocess_labels(y_train)
y_test_01 = preprocess_labels(y_test)
```

Task 4: Neural Network

```
def make_model(input_shape):
```

```
    return model
```

```
model = make_model(input_shape)
```

Task 5: Training

```
def train_model(model):
```

```
    return model
```

```
model = train_model(model)
```

Task 6a: Loss & Metric Curves

```
def plot_curves(history)
```

```
    return fig
```

```
fig = plot_curves(model.history)
fig.show()
```

Task 6b: Figure Caption

[Double-click this cell to write your figure caption in this cell.]

Task 7a: Confusion Matrix

```
def plot_confusion_matrix(x, y, model):  
      
    return fig  
  
fig = plot_confusion_matrix(x_test_scaled, y_test_01, model)  
fig.show()
```

Task 7b: Figure Caption

[Double-click this cell to write your figure caption in this cell.]

Part 2: Short Response (60 points total, 10 points per question)

Question 1: What is a model?

Write your response here:

Question 2: What are the two main types of machine learning we talked about? What is the difference between them?

Write your response here:

Question 3: What is model bias? How is it introduced into our models?

Write your response here:

Question 4: Why do we split our dataset into three separate sets? What are these three sets called?

Write your response here:

Question 5: What is the difference between a model parameter and a hyperparameter?

Write your response here:

Question 6: Why do we (data scientists) make visualizations?

Write your response here: